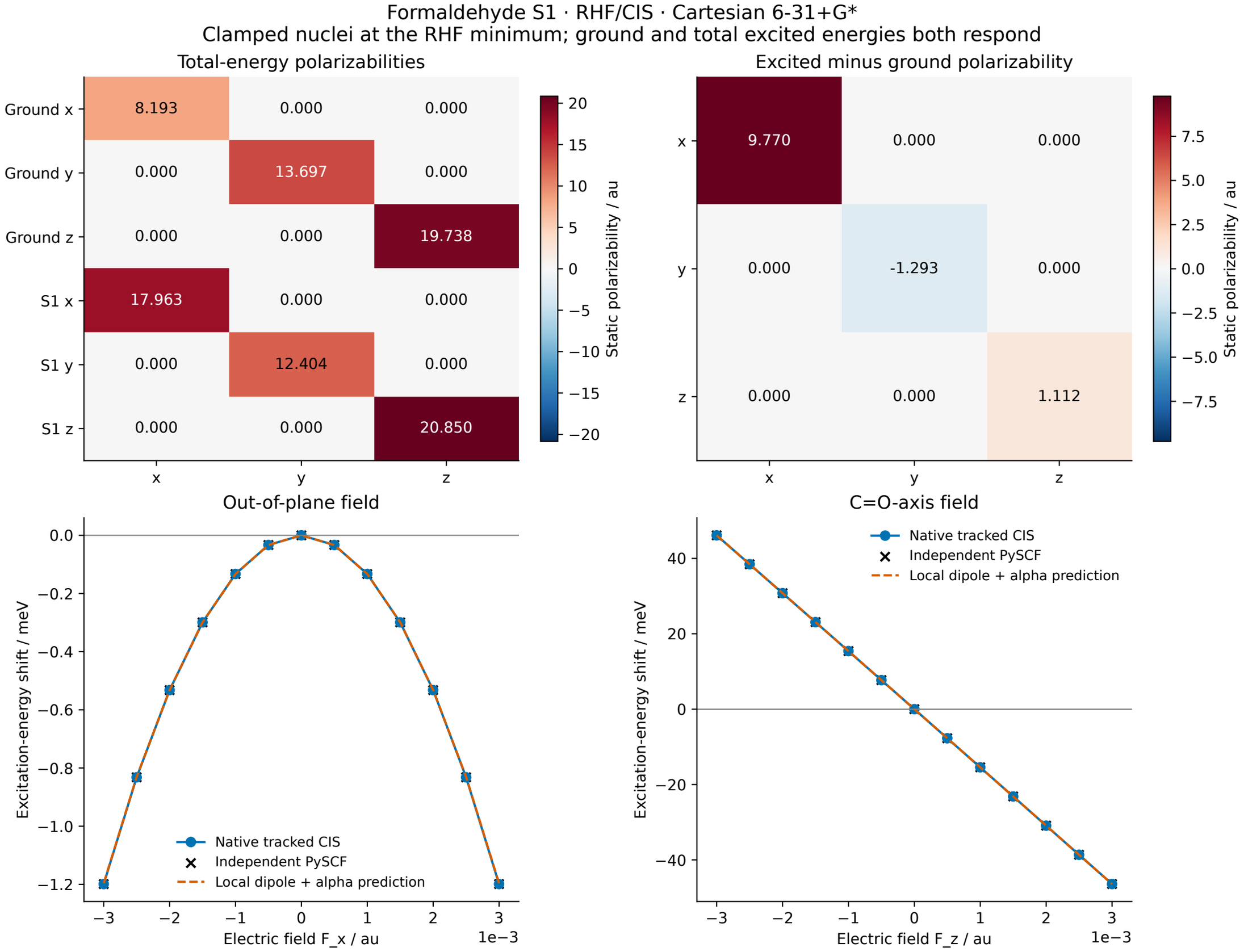
Formaldehyde S1 · RHF/CIS · Cartesian 6-31+G*
Clamped nuclei at the RHF minimum; ground and total excited energies both respond
Total-energy polarizabilities
Ground x
Ground y
Ground z
S1 x
S1 y
S1 z
8.193
0.000
0.000
0.000
13.697
0.000
0.000
0.000
19.738
17.963
0.000
0.000
0.000
12.404
0.000
0.000
0.000
20.850
x
y
z
Static polarizability / au
Excited minus ground polarizability
9.770
0.000
0.000
0.000
-1.293
0.000
0.000
0.000
1.112
Out-of-plane field
Excitation-energy shift / meV
Native tracked CIS
Independent PySCF
Local dipole + alpha prediction
Electric field F_x / au
1e−3
C=O-axis field
Electric field F_z / au